

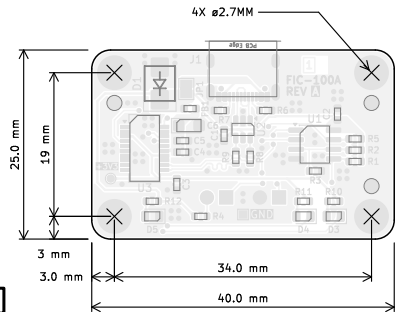
FIC-100A FABRICATION DOCUMENT

LAYER STACKUP LEGEND

	Material	Layer	Thickness	Dielectric	Type	Gerber
	F,Paste				Paste Mask	
	F,Silkscreen				Legend	GBR
	F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
	Core		1,49mm	FR4_7628	Dielectric	
	Copper	L2 (Sig, GND)	0,035mm (1,00oz)		Plane	GBR
	B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	B,Silkscreen				Legend	GBR
	B,Paste				Paste Mask	

Total thickness: 1,6mm
Note: external layer thicknesses are specified after plating

TOP FABRICATION (SCALE 1:1)



IMPEDANCE TABLE

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
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- FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)
- 1) FABRICATE PER IPC-6012A CLASS 2.
 - 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
 - 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
 - 4) SURFACE FINISH: ENIG
 - 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
 - 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
 - 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
 - 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
 - 9) PCB MATERIAL REQUIREMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - B. Tg 170 C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY RYAN DYNAMICS.
 - 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	40,000 × 25,000 mm
BOARD THICKNESS	1,600 mm
TRACE WIDTH	0,200 mm
TRACE TO TRACE	0,200 mm
MIN. HOLE (PTH)	0,300 mm
MIN. HOLE (NPTH)	0,650 mm
ANNULAR RING	0,150 mm
COPPER TO HOLE	0,254 mm
COPPER TO EDGE	0,250 mm
HOLE TO HOLE	0,254 mm

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

PROJECT
USB IR ADPT

DRN
2026-04-24 R. HICKS

CHK

ENG APP

MFR APP

SHEET PATH

FILENAME
FIC-100A.kicad_pcb



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FIC-100A

GIT HASH

d124fa5

DRAWING No

FABRICATION

A4

VARIANT

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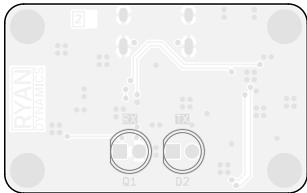
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SHEET

1 OF 8

FIC-100A FABRICATION DOCUMENT

BOTTOM FABRICATION (SCALE 1:1)



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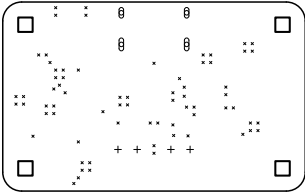
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2026-04-24	R. HICKS	FIC-100A	
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MFR APP		FABRICATION	
SHEET PATH	GIT HASH		
	d124fa5	DRAWING No	A4
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		REVISION	01+ (Unreleased)
		SHEET	2 OF 8

FIC-100A FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	70	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
○	4	0.60mm (23.62mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
+	4	0.90mm (35.43mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
□	4	2.70mm (106.30mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
Total 82						

DRILL DRAWING L1 - L2 (SCALE 1:1)



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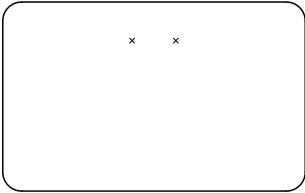
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FIC-100A FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	0.65mm (25.59mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
	Total 2					

DRILL DRAWING L1 - L2 (SCALE 1:1)



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VARIANT

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SHEET

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A

B



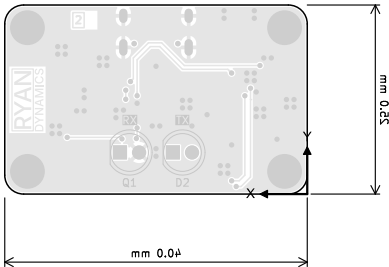
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FIC-100A FABRICATION DOCUMENT

BOTTOM TEST POINTS (SCALE 1:1)

Ref.	Net	X [mm]	Y [mm]
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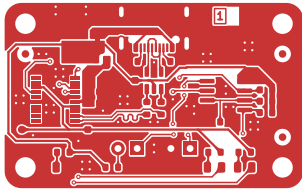


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SHEET PATH	GIT HASH	DRAWING No	A4
	d124fa5		
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FIC-100A.kicad_pcb		1.0+ (Unreleased)	6 OF 8

FIC-100A FABRICATION DOCUMENT

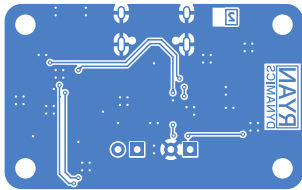
L1 (Sig, PWR) (SCALE 1:1)



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CHK			
ENG APP			
MFR APP		FABRICATION	
SHEET PATH	GIT HASH		
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FIC-100A FABRICATION DOCUMENT

L2 (Sig, GND) (SCALE 1:1)



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